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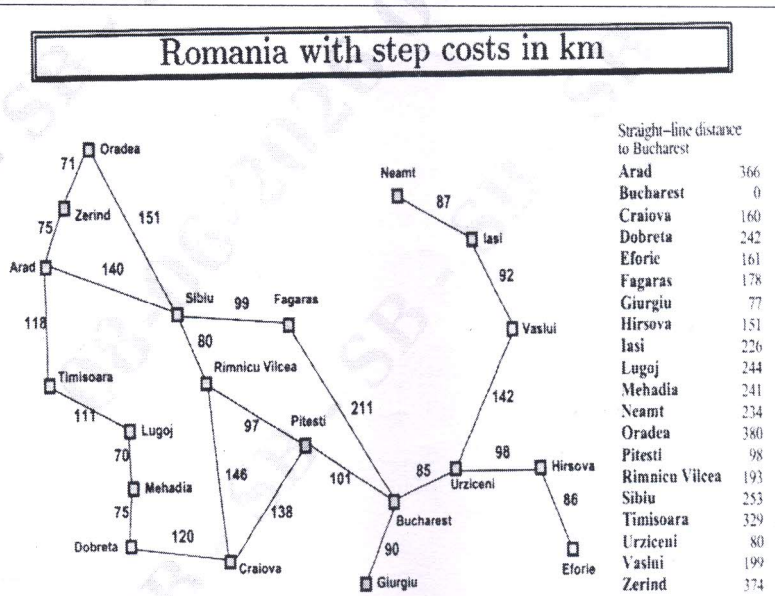
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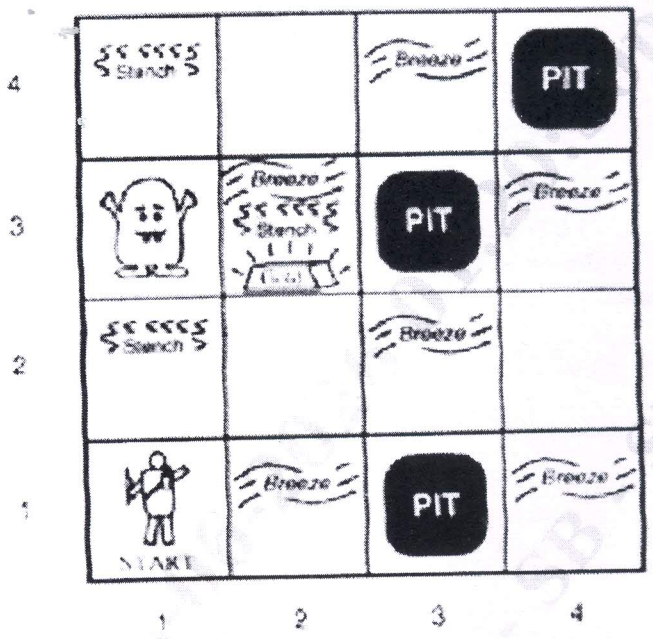
Fifth Semester B.E./B.Tech. Degree Examination, June/July 2026
Artificial Intelligence

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain the state of art of AI.	8	L2	CO1
	b.	Evaluate different types of environment and their impact on the design of intelligent agents.	12	L5	CO1
		OR			
Q.2	a.	Discuss on the types of Intelligent agents with diagram.	16	L1	CO1
	b.	Explain the concept of Rationality with an example.	4	L2	CO1
		Module – 2			
Q.3	a.	Describe how a problem-solving agent approaches a well defined problem suck as the vaccum cleaner.	10	L2	CO2
	b.	Given the 8-Queens problem, apply Breadth First Search (BFS) strategy to find the solution. Draft diagram to illustrate the search tree.	10	L3	CO2
		OR			
Q.4	a.	Define Uninformed Search. List different types of uninformed search.	4	L2 L1	CO2
	b.	Explain any three (3) uninformed search.	16	L2 L1	CO2
		Module – 3			
Q.5	a.	 Fig. Q5 Find the cost using informed search methods like i) Greedy Best First Search and ii) Simple Memory Bounded Heuristic Search (SMA*).	10	L3	CO3
	b.	Explain A* search and its optimality with given map illustration in Fig. Q5.	10	L3	CO3

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		OR			
Q.6		 Fig. Q6 Use / Utilize the Wumpus world grid to generate propositional logic by finding the syntax, semantics, simple knowledge and inference procedure.	**20**	**L3**	**CO3**
		Module – 4			
Q.7	**a.**	Apply the first order logic representation with syntax, semantics..	**10**	**L3**	**CO3**
	b.	Apply the first order logic representation knowledge engineering with appropriate examples.	**10**	**L3**	**CO3**
		OR			
Q.8	**a.**	Differentiate Propositional logic Vs First order logic.	**4**	**L3**	**CO3**
	b.	Write the following sentences in first order logic and prove "John like Bread" using Forward chaining. John likes all kinds of food. Apples are Food. Chicken is Food. Anything anyone eats and isn't killed alive Sue eats everything Bill eats Bread is Food.	**16**	**L3**	**CO3**
		Module – 5			
Q.9	**a.**	Explain Backward Chaining Algorithm and prove that with an example.	**10**	**L2**	**CO4**
	b.	Derive a simple statement using resolution. "If it rains, the ground will be wet"	**10**	**L4**	**CO4**
		OR			
Q.10	**a.**	Write notes on planning graphs in detail.	**10**	**L2**	**CO5**
	b.	Give an account on the algorithms for planning as state-space search.	**10**	**L2**	**CO5**

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026
Artificial Intelligence

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain Alan Turing's significant contribution to artificial intelligence and give a brief introduction to the Turing Test in AI.	10	L2	CO1
	b.	Define agent, agent function and agent program. Explain with a neat diagram how agent interacts with environment through sensors and actuators.	10	L1	CO1
		OR			
Q.2	a.	List the types of Agents. Explain Goal Based and utility based agent with neat diagram.	10	L2	CO1
	b.	Compare and contrast between i) Deterministic and Stochastic ii) Static and Dynamic iii) Episodic and Sequential iv) Fully observable and partially observable. Give example for each of the nature of environment given above.	10	L2	CO1
		Module – 2			
Q.3	a.	Explain the tree search and graph search algorithms.	10	L2	CO2
	b.	Explain problems solving agents along with algorithm and illustrate the incremental formulation of 8-Queens problem.	10	L2	CO2
		OR			
Q.4	a.	List and explain the criteria to measure the performance of search strategies.	10	L2	CO2
	b.	Explain Breadth first search technique as a problem solving strategy with its benefits and shortcomings.	10	L2	CO2
		Module – 3			
Q.5	a.	Explain A* algorithm for shortest path and apply the same for the below graph.	10	L3	CO3

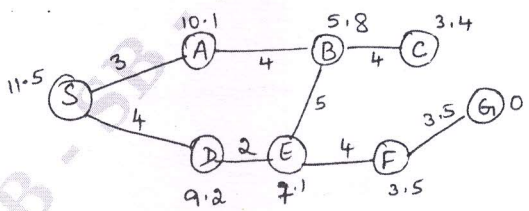
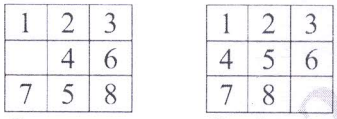
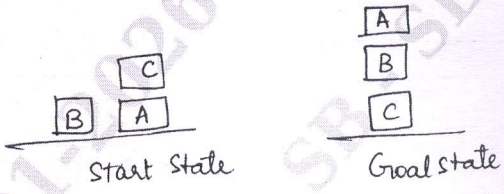


Fig Q5(a)

	b.	Apply heuristic search algorithm on the given 8 puzzle problem to reach the goal state from the initial state  	10	L3	CO3
		OR			
Q.6	a.	Define knowledge based agent. Outline the knowledge based agent program.	10	L1	CO3
	b.	Define Propositional Logic. Explain syntax and semantics.	10	L1	CO3
		Module – 4			
Q.7	a.	Explain first order logic with its syntax in BNF form.	10	L2	CO4
	b.	Explain Quantifiers. Differentiate between Universal and Existential Quantifier.	10	L2	CO4
		OR			
Q.8	a.	Illustrate Kinship Domain with an example.	10	L2	CO4
	b.	Illustrate unification algorithm used for reasoning with example.	10	L2	CO4
		Module – 5			
Q.9	a.	Outline the backward chaining algorithm for definite clauses. Construct a proof tree to prove that "west is a criminal".	10	L2	CO5
	b.	Apply Resolution for "west is a criminal" and "curiosity killed the cat" example.	10	L3	CO5
		OR			
Q.10	a.	Define Planning. Explain block world problem for the following start state and End state.  	10	L2	CO5
	b.	Illustrate how planning graph data structure can be used to give a better heuristic for a planning problem.	10	L2	CO5

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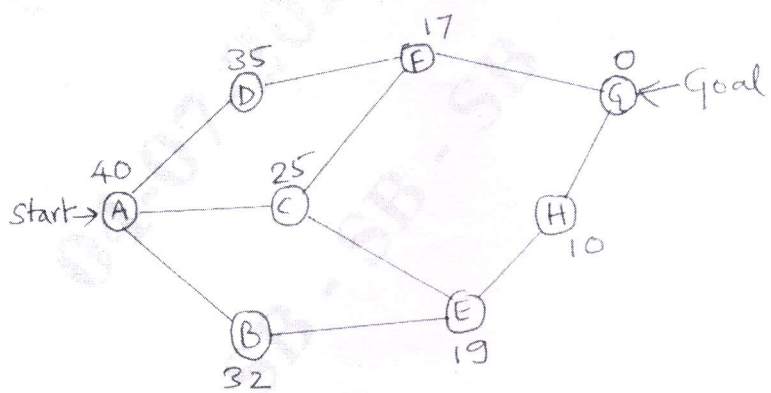
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BCS515B

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025
Artificial Intelligence

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	What is Artificial Intelligence? Explain the four approaches of Artificial Intelligence in detail	10	L2	CO1
	b.	What Artificial Intelligence can do today? Explain various application domains of Artificial Intelligence.	10	L2	CO1
		OR			
Q.2	a.	With the help of a block diagram, explain how agents interact with task environments.	05	L2	CO1
	b.	Briefly explain the properties of task environments.	07	L2	CO1
	c.	With a neat diagram explain the following : i) Utility based agents ii) Learning agents	08	L2	CO1
		Module – 2			
Q.3	a.	List and define the four phases of problem – solving process.	05	L3	CO2
	b.	Explain how a search problem can be formally defined. Illustrate the same for the Vacuum world problem.	10	L2	CO2
	c.	Write the state-space graph for the two-cell vacuum world problem.	05	L3	CO2
		OR			
Q.4	a.	Illustrate Breadth-first search and Depth-first search strategies with suitable example.	10	L3	CO2
	b.	Explain the different metrics used to evaluate an algorithms performance.	06	L2	CO2
	c.	List and explain the three kind of queues used in Best –first search method.	04	L2	CO2
		Module – 3			
Q.5	a.	Apply Greedy best-first search algorithm to the following graph and show the various stages in computing the solution tree.	10	L3	CO3
		 Fig. Q.5 (a)			
	b.	Explain different methods of deriving heuristics.	10	L2	CO3
		OR			
Q.6	a.	Using PEAS description, represent the Wumpus World problem.	06	L3	CO3
	b.	Briefly explain Knowledge Based Agents.	04	L2	CO3
	c.	Explain in detail, syntax and semantics of propositional logic.	10	L3	CO3

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		Module – 4			
Q.7	**a.**	Explain in detail, syntax of first order logic.	**10**	**L2**	**CO4**
	b.	Explain various steps in the knowledge engineering process.	**10**	**L2**	**CO4**
		OR			
Q.8	**a.**	Write Unification algorithm. Explain with suitable example.	**10**	**L2**	**CO4**
	b.	Explain simple forward – chaining algorithm with an example.	**10**	**L2**	**CO4**
		Module – 5			
Q.9	**a.**	Explain Backward – chaining algorithm with an example.	**10**	**L2**	**CO5**
	b.	Illustrate the procedure to convert a sentence of first order logic to Conjunctive Normal Form (CNF)	**10**	**L2**	**CO5**
		OR			
Q.10	**a.**	Illustrate categories and objects with suitable examples.	**10**	**L2**	**CO5**
	b.	Explain the following with respect to events: i) Time ii) Fluents and objects	**10**	**L2**	**CO5**

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BCS515B

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Artificial Intelligence

Time: 3 hrs. Max. Marks: 100

Note: **1.** ***Answer any FIVE full questions, choosing ONE full question from each module.***
2. ***M : Marks , L: Bloom's level , C: Course outcomes.***

		Module – 1	M	L	C
Q.1	a.	Define the following : i) Intelligence ii) Artificial Intelligence iii) Agent iv) Rationality v) Logical reasoning.	5	L2	CO1
	b.	Examine the AI literature to discover whether the following tasks can currently be solved by computers. i) Playing a decent game of table tennis (ping-pong) ii) Discovering and proving new mathematical theorems iii) Giving competent legal advice in a specialized area of law iv) Performing a complex a surgical operation.	8	L2	CO1
	c.	Implement a simple reflex agent for the vacuum environment. Run the environment with this agent for all possible initial dirt configurations and agent locations. Record the performance score for each configuration and the overall score.	7	L3	CO1
		OR			
Q.2	a.	Is AI a science, or is it engineering or neither or both? Explain.	5	L2	CO1
	b.	Write pseudocode agent programs for the goal based and utility based agents.	8	L1	CO1
	c.	For each the following activities give a PEAS description. i) Playing a tennis match ii) Performing a high jump iii) Bidding on an item in an auction.	7	L1	CO1
		Module – 2			
Q.3	a.	Explain why problem formulation must follow goal transformation.	5	L1	CO1
	b.	Give complete problem formulation for each of the following choose a formulation that is precise enough to be implemented. i) Using only four colors, you have to color a planar graph in such a way that no two adjacent regions have the same color. ii) A 3 – foot – tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get the bananas. The room contains two stackable, moveable, climbable 3-foot high crates.	8	L2	CO2
	c.	Prove each of the following statements or given counter example : i) Breadth – first search is a special case of uniform – cost search. ii) Uniform – cost search is a special case of A* search.	7	L2	CO2

		OR			
Q.4	**a.**	Define the following terms with example. i) State space ii) Search node iii) Transition model iv) Branching factor.	**8**	**L2**	**C**
	b.	Show that the 8-puzzle states are divided in to two disjoint sets, such that any state is reachable from any other state in the same set, while no state is reachable from any state in the other set. Devise a procedure to decide which set a given state is in and explain why this is useful for generating random state.	**7**	**L2**	**CO2**
	c.	Describe a state space in which iterative deepening search performs much worse than depth first search for example, $O(n^2)$Vs $O(n)$).	**5**	**L2**	**CO2**
		Module – 3			
Q.5	**a.**	Devise a state space in which A^* using GRAPH-SEARCH returns a suboptimal solution with h(n) function that is admissible but inconsistent.	**7**	**L2**	**CO3**
	b.	Which of the following are correct? i) $(A \vee B) \wedge (\neg C \vee \neg D \vee E) F (A \vee B)$ ii) $(A \vee B) \wedge (\neg C \vee \neg D \vee E) F (A \vee B) \wedge (\neg D \vee E)$ iii) $(A \vee B) \wedge \neg (A \Rightarrow B)$ is satistiable iv) $(A \Leftrightarrow B) \Leftrightarrow C$ has the same number of models as $(A \Leftrightarrow B)$	**8**	**L1**	**CO3**
	c.	Consider a vocabulary with only four propositions, A, B, C and D. How many models are there for the following sentences? i) $B \vee C$ ii) $\neg A \vee \neg B \vee \neg C \vee \neg D$ iii) $(A \Rightarrow B) \wedge A \wedge \neg B \wedge C \wedge D$.	**5**	**L1**	**CO3**
		OR			
Q.6	**a.**	Prove that if a heuristic is consistent, it must be admissible. Construct an admissible heuristic that is not consistent.	**8**	**L1**	**CO3**
	b.	Prove each of the following assertions : i) $\alpha \equiv \beta$ if and only if the sentence $(\alpha \Leftrightarrow \beta)$ is valid ii) $\alpha \neq \beta$ if and only if the sentence $\alpha \wedge \neg \beta)$ is unsatisfiable.	**7**	**L1**	**CO3**
	c.	Prove, or find a counter example to each of the following assertions. i) If $\alpha \neq (\beta \wedge \gamma)$ then $\alpha \neq \beta$ and $\alpha \neq \gamma$ ii) If $\alpha \neq (\beta \vee \gamma)$ then $\alpha \neq \beta$ and $\alpha \neq \gamma$ (or) both	**5**	**L1**	**CO3**
		Module – 4			
Q.7	**a.**	Which of the following are valid necessary true sentences? i) $(\exists x\, x = x) \Rightarrow (\forall y \exists z\, y = z)$ ii) $\forall x\, P(x) \vee \neg p(x)$ iii) $\forall x$ smart$(x) \vee (x = x)$	**7**	**L1**	**CO4**
	b.	Prove that universal Instantiation is sound that existential instanticition produces an inferentially equivalent knowledge base.	**5**	**L1**	**CO4**

	c.	Write down logical representations for the following sentences, suitable for use with generalized modulus ponens : i) Horses, cows and pigs are mammals ii) Bluebeard is Charlie's parent iii) Offspring and parent are inverse relations	8	L1	CO4
		OR			
Q.8	**a.**	Consider a knowledge base containing just two sentence ; P(a) and P(b) does this knowledge base entail $\forall x\, P(x)$? Explain your answer interms of models.	5	L2	CO4
	b.	Suppose a knowledge base contains just one sentence, $\exists x$ AsHigh As(x, Everest) which of the following are legitimate results of applying existential instantiation? i) AsHighAs(Kilimanjaro, Everest) ii) AsHighAs(Kilimanjaro, Everest) $\wedge$ AsHighAs (Benvevis, Everest)	8	L2	CO4
	c.	Explain how to write any 3-SAT problem of arbitrary size using a single first order definite clause and no more than 30 ground facts.	7	L2	CO4
		Module – 5			
Q.9	**a.**	i) Give a backward chaining proof of the sentence $7 \leq 3 + 9$. Show only the steps that leads to success ii) Give a forward chaining proof of the sentence $7 \leq 3 + 9$. Show only the steps that leads to success.	8	L1	CO5
	b.	Describe the differences and similarities between problem solving and planning.	5	L2	CO5
	c.	Prove that backward search with PDDL problems is complete.	7	L1	CO5
		OR			
Q.10	**a.**	The following prolog code defines a predicate P P(x, [x|y]), P(x, [y|z]) :- P(x, z) i) Show proof trees and solutions for the queries P(A, [2, 1, 3]) and P(z,[1, A, 3]) ii) What standard list operation does P represent?	8	L1	CO5
	b.	Explain why dropping negative effects from every action schema in a planning problem results in a relaxed problems.	5	L2	CO5
	c.	Prove the following assertions about planning graphs : i) A literal that does not appear in the final level of the graph connot be achieved. ii) The level cost of a literal in a serial graph is no greater than the actual cost of an optimal plan for achieving it.	7	L1	CO5

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